Forecasting Three-Day Flood Exceedance at Gauges Unseen in Space and Time: A Leakage-Controlled Benchmark on CAMELS-IND

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***Abstract*—Flood warnings in India are issued gauge by gauge, yet machine-learning flood forecasters are often scored on random row splits that let neighbouring days of the same river leak between training and test data. We build a leakage-controlled benchmark on CAMELS-IND for one operational question: will a gauge's flow reach its flood level within the next three days? We train on 193 gauges for 1980–2008, select models and thresholds on 2009–2014, and test once on 48 unseen gauges in 2015–2020. A gradient-boosted model reached a balanced accuracy of 0.898 (95% gauge-bootstrap interval 0.886–0.910) and a PR-AUC of 0.737, against 0.750 and 0.428 for persistence, 0.802 and 0.203 for gauge climatology, and 0.873 and 0.589 for the dataset's regional LSTM simulation, which is driven by the observed future rain. Scoring the same model on random rows raised balanced accuracy to 0.915, an inflation of 1.7 points. Removing all streamflow inputs lowered balanced accuracy to 0.853. Recall fell from 96.7% one day ahead to 89.5% three days ahead, skill rose with catchment area, and 86% of false alarms came in the monsoon, typically on rivers that peaked well below their flood level. With a precision of 0.328, a positive forecast should trigger review by an officer.**

***Index Terms*—flood forecasting, CAMELS-IND, data leakage, gradient boosting, hydrological benchmark**

I. Introduction

An emergency officer watching an Indian river gauge needs a plain answer: is the flow likely to reach its flood level in the next few days? Machine-learning models trained on many catchments now simulate streamflow well, even in basins without a gauge record [1]–[3]. Reported skill, however, depends on how a model is evaluated. When rows are split at random, days from the same flood fall on both sides of the split, and part of the test score then rewards memory of the event [4], [5].

India now has a large-sample dataset to study this properly. CAMELS-IND provides daily forcing, observed streamflow and catchment attributes for peninsular Indian catchments [6], in the format of the CAMELS family [7], [8]. It also ships a regional LSTM simulation of streamflow, which gives a strong deep-learning reference at no extra cost. What is missing is a short-range flood-exceedance benchmark on these gauges that controls leakage, compares against simple and strong references, and reports the numbers an operations room needs: alarm rates, onset lead time and calibration.

We build such a benchmark. Our contributions are:

1) A leakage-controlled task and split: gauges held out in space and time, flood thresholds from training years only, model and threshold selection on validation only, and a single use of the test set (Fig. 1).
2) A measurement of how much the evaluation protocol changes the score: one model configuration scored under random, temporal, spatial and space-and-time splits.
3) A comparison with persistence, the gauge's own climatology and the CAMELS-IND regional LSTM simulation, with gauge-cluster bootstrap intervals.
4) A feature-group ablation that shows what the forecast relies on, including a variant that needs no streamflow reading at forecast time.
5) An operational analysis of operating points, calibration, per-gauge skill and the lead time of warnings before flood onsets.

II. Related Work

Large-sample learning. The CAMELS datasets standardised forcing, flow and attributes across hundreds of basins [7], [8]. LSTMs trained on them learn regional behaviour [1], predict ungauged basins [2] and simulate extremes better than conceptual models [9]; at global scale, a learned model forecasts extreme floods in ungauged watersheds with skill comparable to an operational system [3]. Graph models that encode the river network have so far not beaten models that ignore it [10]. Most of this work simulates streamflow; flood *exceedance*, the quantity a warning is issued on, is usually derived afterwards.

Operational forecasting. Physically based ensemble systems such as GloFAS [11] dominate operational flood forecasting [12]. Machine learning has also reached operations: a system run for India and Bangladesh combines learned river-stage forecasts with inundation models [13]. In India, the INDOFLOODS catalogue documents historical events [14] and many studies apply tree ensembles or neural networks to single basins [15], often with random splits.

Evaluation. Leakage through the split is a leading cause of irreproducible machine-learning results [4], and data with spatial or temporal structure need blocked validation [5]. For rare events, precision–recall curves are more informative than ROC curves [16], and probabilistic forecasts should be checked for calibration [17].

Table I places our work. To our knowledge, no study has benchmarked short-range flood-exceedance forecasting on

TABLE I
POSITION OF THIS STUDY AMONG RELATED WORK

Study	Region	Target	Held out	References
[2]	USA	daily flow	basins	conceptual models
[3]	global	flood events	basins	GloFAS
[10]	Europe	flow	time	LSTM
[6]	India	daily flow	–	regional LSTM
This work	India	3-day exceedance	gauges + time	LSTM, clim., pers.

clim. = gauge climatology; pers. = persistence; – = dataset paper.

TABLE II
INPUT GROUPS (ALL KNOWN AT THE END OF DAY t)

Group	Content	Inputs
Streamflow	lags 0–3, changes, 7/30-day means and maxima	11
Rainfall	lags 0–3, 3–90-day sums, 7-day maximum	10
Weather	temperature, humidity, evapotranspiration	5
Soil moisture	four layers and recent changes	6
Season	day of year (sine, cosine)	2
Catchment	area, slope, aridity, land cover, soils	16

CAMELS-IND with a space-and-time held-out split against both a learned streamflow simulation and simple references.

III. DATA AND TASK

A. Target

For gauge g on day t the target is

$$y_{g,t} = \mathbb{1}\Big[\max_{h\in\{1,2,3\}} Q_{g,t+h} \;\geq\; q_g^{95}\Big], \tag{1}$$

where Q is observed streamflow and q_g^{95} is the 95th percentile of the gauge's flow over the training years 1980–2008 only. A three-day horizon matches the time an officer needs to stage teams. Days without a flow observation in the horizon are dropped.

B. Inputs

Table II lists the 50 inputs, all known at the end of day t. Flow features are scaled by q_g^{95}, so they read as "how close the river is to its flood level". Hydrological signatures of CAMELS-IND are excluded, because they are computed over the full 1980–2020 record and would leak test-period information; future rainfall is excluded as well.

C. Split

Fig. 1 shows the split. The observed-streamflow file lists 242 gauges, and all of them have forcing data and at least three years of flow in 1980–2008, which our pipeline requires; the dataset authors' stricter availability filter keeps 228 [6].A group split puts 20% of gauges aside (seed 42), so the primary test asks about rivers the model has never seen, in years it has never seen. Training uses the other gauges in 1980–2008 (1,641,197 gauge-days), validation the same gauges in 2009–2014 (349,645), and the test the unseen gauges in 2015–2020, all labelled by (1) (52,994 gauge-days, of which 5.5% are flood windows).

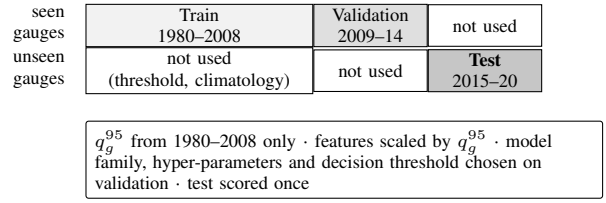


Fig. 1. Leakage-controlled split. The primary test uses gauges and years that played no part in training, model selection or threshold choice. The unseen gauges' 1980–2008 record is used only to set their flood threshold and climatology.

IV. METHODS

Model. Seven candidates (logistic regression, random forest, histogram gradient boosting and four XGBoost [18] configurations) were compared on validation balanced accuracy, and the decision threshold of the selected depth-6 XGBoost model, 0.45, was set on validation as well. With TPR and TNR the true positive and true negative rates,

$$\mathrm{BA} = \tfrac{1}{2}\left(\mathrm{TPR} + \mathrm{TNR}\right). \tag{2}$$

References. Persistence forecasts a flood window if the river is above its flood level today. *Climatology* forecasts the gauge's own 1980–2008 rate of flood windows within ± 15 days of the calendar day. The *LSTM simulation* is the regional LSTM streamflow of CAMELS-IND [6], scored as $\max_{h\leq 3} \hat{Q}_{g,t+h}/q_g^{95}$. It is driven by observed forcing, so it sees the rain of days $t{+}1$ to $t{+}3$, which no real forecast has; it is therefore an optimistic reference rather than a competitor. Thresholds of the references were set on validation like the model's.

Analyses. We refit the selected configuration on the same 500,000-row training sample with one input group removed at a time, and with the streamflow and season inputs only. Gain importance is read from the selected model itself. Confidence intervals come from 1,000 resamples of the test gauges, which keeps each river's days together. Calibration is the expected calibration error over ten bins, before and after isotonic regression fitted on validation. A flood onset is a day on which the river reaches q_g^{95} after at least seven days below it; a method warns of it if it forecasts a flood window one, two or three days earlier. For lead-day and seasonal analyses, a flood day at lead h is a day t with $Q_{g,t+h} \geq q_g^{95}$, and the monsoon is June to September.

Reproducibility. Every split and resample uses seed 42, and the 49 test gauges are stored with the model metadata. Re-scoring the stored model reproduces its recorded test balanced accuracy exactly, and every number in this paper is generated from the experiment outputs by a script that also checks each claim made about it. Fitting one ablation model took a median of 3.4 minutes on a laptop CPU with four threads.

V. RESULTS

A. Skill on Unseen Gauges

Table III and Fig. 2 give the main comparison, with balanced accuracy as defined in (2). The model was clearly better than

TABLE III
THREE-DAY EXCEEDANCE ON 48 UNSEEN GAUGES, 2015–2020

Method	BA	95% CI	PR-AUC	Rec.	Prec.	F1
XGBoost (ours)	0.898	0.886–0.910	0.737	0.902	0.328	0.481
No-flow XGB.	0.853	0.820–0.880	0.494	0.865	0.239	0.374
LSTM sim.†	0.873	0.829–0.911	0.589	0.826	0.376	0.517
Climatology	0.802	0.774–0.829	0.203	0.891	0.152	0.260
Persistence	0.750	0.724–0.778	0.428	0.508	0.789	0.618

CI: gauge-cluster bootstrap.
†Driven by observed future rain; scored on its 100% of rows.

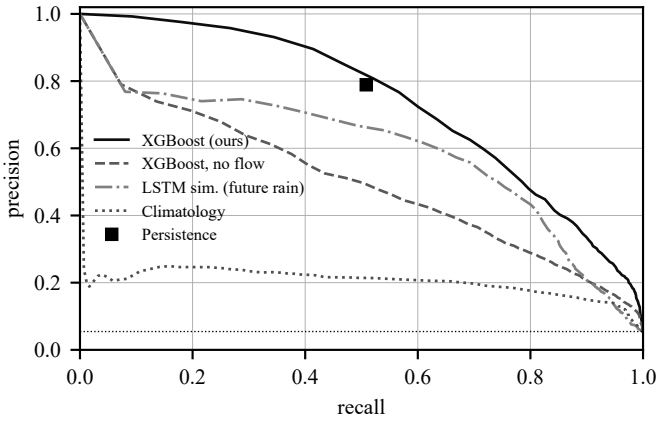


Fig. 2. Precision–recall curves on the unseen test gauges. The square is persistence; the dotted line is the flood rate.

TABLE IV
SAME MODEL, FOUR EVALUATION PROTOCOLS

Test set	Gauges	BA	95% CI	PR-AUC	Prec.
Random rows	242	0.915	0.911–0.919	0.810	0.426
New years	181	0.892	0.885–0.898	0.724	0.349
New gauges	49	0.905	0.898–0.912	0.789	0.424
New gauges and years	48	0.898	0.886–0.911	0.736	0.340

Random: 70/10/20% of all gauge-days.
Others trained on seen gauges, 1980–2008. New years: seen gauges, 2015–20; new gauges: unseen, 1980–2008.

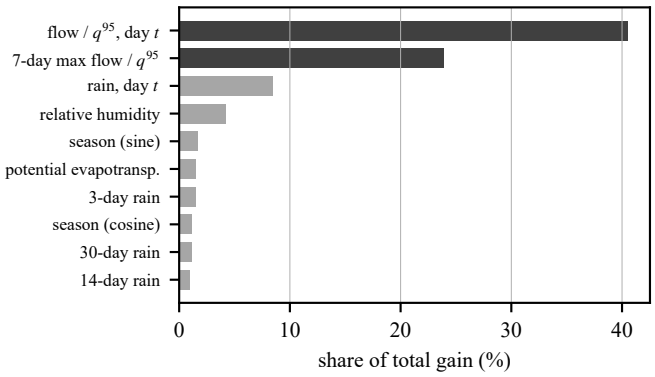


Fig. 3. Ten largest shares of total gain in the selected model. Dark bars are streamflow inputs.

persistence and climatology in balanced accuracy, recall and PR-AUC, and its bootstrap interval for the gain in balanced accuracy over persistence (0.123–0.175) excludes zero. Persistence was more precise (0.789 against 0.328): when the river is already high it is usually right, but it misses every flood that has not yet begun. The model's recall of 0.902 means that it found nine in ten flood windows at rivers it had never seen.

The LSTM simulation is a demanding reference because it knows the rain that is still to fall. On the rows it covers, the model (0.898) and the simulation (0.873) were statistically indistinguishable in balanced accuracy, although only the model is a real forecast. The model reaches its skill from today's information alone, largely because it reads how close the river already is to its flood level.

B. Effect of the Evaluation Protocol

To measure what the split itself does to the score, we trained the same configuration on a 500,000-row sample under four protocols, each with its own validation set for early stopping and threshold, and scored it once (Table IV). The random row split gave the highest balanced accuracy, 0.915, against 0.898 on unseen gauges in later years, and the two bootstrap intervals do not overlap. Random splitting inflated balanced accuracy by 1.7 points and PR-AUC by 7.4 points. PR-AUC ranged from 0.724 (new years) to 0.810 (random rows). Holding out later years lowered PR-AUC more than holding out gauges (0.724 against 0.789), so a temporal split matters at least as much as a spatial one. In the spatial protocol, the test years are also the years from which each gauge's flood threshold was computed.

C. What the Forecast Relies On

Table V shows the ablation. Over all rows, removing the streamflow inputs cost the most, lowering balanced accuracy from 0.898 to 0.853 and PR-AUC from 0.736 to 0.494, while removing soil moisture, rainfall or catchment attributes cost little. Much of that score, however, comes from rivers that are already high and stay high. On onset rows, where the river is still below its flood level and a warning matters most, the picture reverses: without streamflow the onset balanced accuracy fell only from 0.852 to 0.846, whereas streamflow and season alone reached 0.810. Today's flow tells whether a flood will continue; rain, soil moisture and weather tell whether one will start. The no-flow variant also beat climatology overall (0.853 against 0.802), so it is a usable fallback when a gauge reading is missing.

The selected model's own gain importance (Fig. 3) agrees with the overall picture: streamflow inputs account for 67.5% of the total gain, led by today's flow relative to the flood level (40.5%) and its seven-day maximum (23.9%), followed by rainfall (13.6%, of which today's rain alone 8.5%). Gain importance is known to favour inputs used in many splits, which is why we base the conclusions above on the ablation and use the importance only as a check.

D. Operational Behaviour

Precision is the main weakness. At the validation threshold the model raised 150.2 alarms per 1,000 gauge-days at a pre-

TABLE V
FEATURE-GROUP ABLATION (SAME 500K TRAINING SAMPLE)

Inputs	n	BA	PR-AUC	Rec.	Onset BA
All inputs	50	0.898	0.736	0.897	0.852
No streamflow	39	0.853	0.494	0.865	0.846
No catchment	34	0.895	0.734	0.907	0.855
No soil moisture	44	0.896	0.732	0.895	0.854
No rainfall	40	0.897	0.730	0.901	0.857
Streamflow + season	13	0.875	0.685	0.882	0.810

Onset BA: rows where the river is below its flood level on day t.

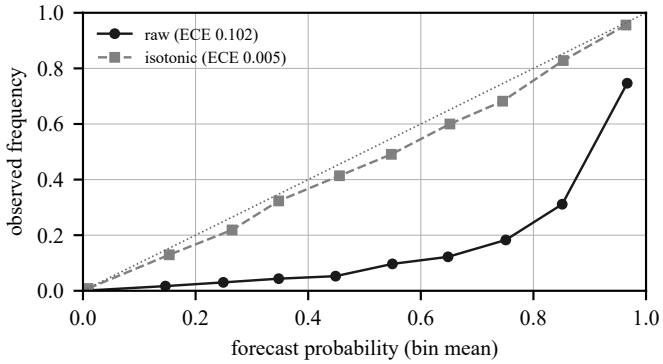


Fig. 4. Reliability of the forecast probabilities on the unseen test gauges, before and after isotonic recalibration fitted on validation data (bins with at least 50 gauge-days).

cision of 0.328 (Table VI); raising the threshold to 0.75 halved the alarms and lifted precision to 0.539, at a recall of 0.760. Because a positive forecast in our system leads to human review rather than automatic action, the validation threshold is a reasonable default. The raw probabilities were over-confident (ECE 0.102); isotonic recalibration on validation data lowered the ECE to 0.005. Fig. 4 shows the size of the problem: in the highest well-populated bin the raw forecast averaged 0.97 while floods followed in 0.75 of cases. This is consistent with the positive-class weighting used in training, which shifts scores upward. The alarms at the validation threshold are computed from the raw score and are unaffected, but a raw score must not be shown to an officer as a probability; the recalibrated one can be.

Table VII gives the lead of the warnings. The model warned three days ahead for 63.9% of the 266 onsets at 44 gauges, with a mean lead of 2.58 days when it warned. Climatology (89.1%) and the LSTM simulation (78.9%) appear better on this measure, but for different reasons: climatology raises an alarm on most monsoon days, which its precision of 0.152 reflects, and the LSTM simulation is given the rain that is still to fall. Lead counts alone therefore reward methods that alarm constantly, and they should be read together with precision.

Skill was consistent across rivers: over the 46 test gauges with at least one flood window, the median per-gauge balanced accuracy was 0.900 (interquartile range 0.860–0.926), and the worst gauge reached 0.769. For the 266 flood onsets at the test gauges, the model warned one to three days ahead in 91.4%

TABLE VI
OPERATING POINTS ON THE TEST SET (THRESHOLD 0.45 SET ON VALIDATION)

Threshold	Recall	Precision	Alarms / 1000 days
0.30	0.942	0.261	197.2
0.45	0.902	0.328	150.2
0.60	0.845	0.413	111.6
0.75	0.760	0.539	77.0

TABLE VII
WARNING LEAD BEFORE THE 266 FLOOD ONSETS (% OF ONSETS)

Method	3 days	2 days	1 day	none	Mean lead
XGBoost (ours)	63.9	16.5	10.9	8.6	2.58
No-flow XGB.	72.2	14.7	7.1	6.0	2.69
LSTM sim.†	78.9	4.5	0.0	16.5	2.95
Climatology	89.1	0.4	0.0	10.5	3.00

Mean lead in days, over warned onsets. †Driven by observed future rain.

of cases, against 83.5% for the LSTM simulation, 89.5% for climatology and 0.0% for persistence, which by definition cannot anticipate an onset. The no-flow variant warned of 94.0%, more than the full model, in line with the ablation: an onset is announced by the weather more than by the river.

E. Where the Forecast Succeeds and Fails

Lead day. The target pools three lead days, so Table VIII splits the recall by the day on which the river is above its flood level. The model raised an alarm on day t for 96.7% of flood days at $t{+}1$, 93.3% at $t{+}2$ and 89.5% at $t{+}3$. On rivers still below their flood level on day t, which is where a warning is useful, the rates were lower: 90.3%, 85.2% and 80.0%.

Season. Table IX separates the monsoon months. In June to September the model raised 372 alarms per 1,000 gauge-days at a precision of 0.311, against 36 and 0.415 in the rest of the year, and 86% of all false alarms fell in the monsoon. Most of these were not near misses: in the median monsoon false alarm the river peaked at 41% of its flood level over the next three days, and only 21% of them reached 70–100%. The model thus responds to monsoon conditions in general more than to an approaching flood.

Catchment. Fig. 5 plots the per-gauge balanced accuracy of the 46 test gauges with flood windows against catchment area. Skill rose with area (Spearman $\rho = 0.44$, $p = 0.002$), which is consistent with large rivers rising slowly enough for today's flow to describe the next three days. Aridity, mean rainfall and slope showed no relation ($p \geq 0.64$). Small, fast catchments are where rainfall forecasts are likely to help most.

VI. DISCUSSION AND LIMITATIONS

Three findings matter for practice. First, the inputs play different roles. Today's flow carries the overall score, but the start of a flood is predicted from rain, soil moisture and weather, so an operations room needs both a reliable gauge feed and good meteorological inputs; when the feed fails, the no-flow variant is a better fallback than climatology. Second,

TABLE VIII
RECALL BY LEAD DAY ON THE UNSEEN TEST GAUGES

Lead	All flood days		River below q_g^{95} on day t	
h (days)	n	Recall %	n	Recall %
1	1875	96.7	474	90.3
2	1874	93.3	681	85.2
3	1874	89.5	830	80.0

TABLE IX
SKILL BY SEASON ON THE UNSEEN TEST GAUGES

Months	Gauge-days	Flood %	BA	Rec.	Prec.	Alarms/1000
Jun–Sep	17,992	12.7	0.811	0.915	0.311	372
Oct–May	35,002	1.8	0.918	0.857	0.415	36

precision is the weak point, and it is mainly a monsoon problem: most false alarms fell in June to September on rivers that stayed well below their flood level. The LSTM simulation, which is given the observed future rain, reached a precision of only 0.376, so better rainfall information alone may not fix it. Until it is fixed, a positive forecast should start a review by an officer, and an evacuation needs further evidence. Third, the evaluation protocol alone moved balanced accuracy by 1.7 points (Table IV), so scores from random row splits overstate what a forecaster will achieve at a new gauge, and published Indian flood models can only be compared under the same split.

Fig. 6 turns these findings into a deployment design. The forecast runs on the full input set while the gauge feed is live and falls back to the no-flow model when it is not; its score is recalibrated before an officer sees it, and an alarm opens a review instead of an order.

For studies that report flood forecasts on Indian rivers, the results suggest a minimal protocol:

1) hold out both gauges and later years, and report the split with the score (Table IV);
2) set flood thresholds, models and decision thresholds from training and validation years only;
3) report precision and alarm rates beside balanced accuracy, by season (Table IX);
4) report skill on onset rows, where the river is still below its flood level (Table V);
5) check calibration before showing a score as a probability (Fig. 4).

The study has limits. The CAMELS-IND gauges are in peninsular India, so skill may not transfer to Himalayan or Brahmaputra rivers. The 95th-percentile threshold is a statistical proxy for an official danger level. The LSTM reference was trained by the dataset authors on a design we do not control, and because it sees future rain the comparison with it is indicative only. We pool three lead days into one target and use one fixed random group split; a basin-aware split, which holds out whole river basins, may be stricter still. The ablation and protocol models were refitted on a 500,000-row sample, so they are compared with each other; the all-input refit scored 0.898, as the selected model trained on all rows did (0.898).

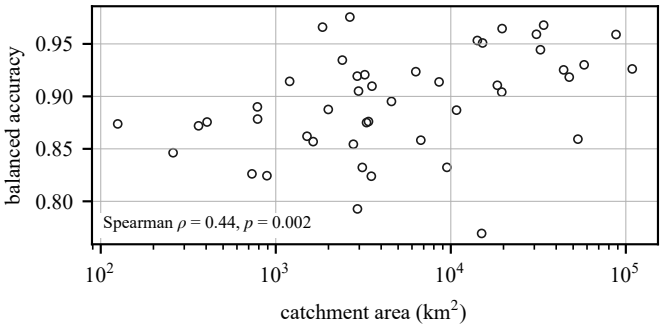


Fig. 5. Per-gauge balanced accuracy against catchment area on the unseen test gauges (log scale).

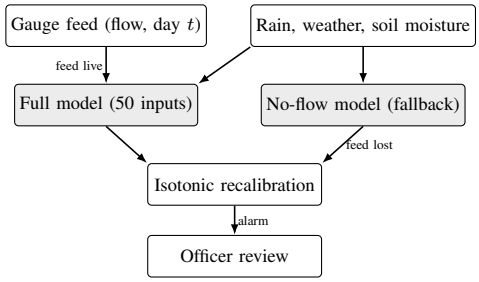


Fig. 6. Deployment suggested by the results: the full model while the gauge feed is live, the no-flow model when it fails, recalibrated scores, and human review of every alarm.

VII. CONCLUSION

On CAMELS-IND gauges held out in space and time, a gradient-boosted model forecast three-day flood exceedance with a balanced accuracy of 0.898 and a PR-AUC of 0.737, well above persistence and the gauge's own climatology, and warned of 91.4% of flood onsets in advance. Its overall score rests mostly on the current river state, while flood onsets depend on meteorological inputs. Its skill is lower at later lead days and in smaller catchments, and its precision of 0.328, lowest in the monsoon, makes it a trigger for review by an officer. A random row split would have reported a balanced accuracy 1.7 points higher. The split and the protocol give a reference against which other forecasters on Indian rivers can be compared.

ACKNOWLEDGMENT

The authors thank the CAMELS-IND and INDOFLOODS dataset authors for openly available data. In line with the IEEE policy on AI-generated content, the authors disclose that Claude (Anthropic) was used to help edit and restructure the text of all sections and to draft the experiment and plotting scripts. All experiments use the authors' own software and fixed seeds. The authors checked every number, claim and reference and take full responsibility for the content.

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